

Title

Project Report

MSc Research Project / (Research) Practicum / Internship
Programme Name

Forename Surname

Student ID: XXX

School of Computing
National College of Ireland

Supervisor: XXX

September 2026

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XXX

*Abstract This project introduces a novel approach to 'Reliability and Recovery of Amazon SQS Messaging under Injected Consumer and Downstream Failures'. We address the gap in Kyrychenko et al. (2025b) where The baseline evaluates SQS standard operations under steady load but misses real-world downstream failures such as consumer crash loops, throttling, and API timeouts. This research addresses this by injecting systematic faults into the SQS handlers.. Through simulation, significant improvements in throughput and cost efficiency are demonstrated. Introduction Cloud computing represents a shift. We investigate 'Reliability and Recovery of Amazon SQS Messaging under Injected Consumer and Downstream Failures'.

In reference to Jackson (2025), the integration highlights a distinct improvement metric.

Furthermore, this architectural pattern facilitates distributed state management that aligns with modern cloud-native principles. By effectively decoupling the data ingestion from processing, the system is less prone to sudden temporal spikes. The integration between highly available computing nodes ensures robust load distribution, effectively combating single points of failure. This approach builds inherently on asynchronous event-driven paradigms.

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[IMAGE PLACEHOLDER: terminal_output.png] *Conclusion* In conclusion, we successfully validated

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